

Kafr El Dawwar City - Baseline 2016 vs Comparison 2026

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Executive Evidence Summary

Study scope: Kafr El Dawwar City (CITY ONLY)

2016 agricultural baseline area: 1.4317 km2

Detected encroachment area: 80,823.53 m2

Detected encroachment area: 19.24 feddan

Share of 2016 agricultural baseline converted: 5.65%

Share of total city area: 1.37%

Relative carbon-potential loss share: 4.44%

Absolute tons CO2 lost: NOT QUANTIFIED

Method and Governance

The result is a satellite-derived change-detection proxy based on Sentinel-2 NDVI/NDBI rules. It is not a legal determination of illegal encroachment. The agricultural baseline is satellite-derived, and cadastral records plus field verification remain required for enforcement. The carbon result is a relative potential-loss percentage only. No absolute tons-CO2 claim is made.

Priority Hotspot

KED-ENCR-0171 - 14,608.5 m2 (3.48 feddan).

Formal Recommendations for Authorities

1. The satellite-derived change-detection result identifies 5.65% of the city's 2016 satellite-derived agricultural baseline as converted to satellite-visible built-up land between the baseline and comparison periods. Field verification of the highest-ranked hotspots is recommended before enforcement action.
2. The largest detected patch should be prioritized for the first field-verification visit because patch size provides a transparent basis for inspection sequencing.
3. The detected encroached area corresponds to an estimated 4.44% of the 2016 agricultural carbon-sequestration potential represented by the notebook's NDVI-normalized relative model. This is a relative potential-loss share, not an absolute carbon-stock or tons-CO2 measurement.
4. This analysis is a satellite-derived change-detection proxy and does not constitute a legal or administrative determination of illegal encroachment. Field verification and cadastral cross-checking are required before enforcement.

Evidence and Output Files

/content/drive/MyDrive/ENVA/Agricultural_Encroachment/evidence/consolidated_evidence_ledger.json
/content/drive/MyDrive/ENVA/Agricultural_Encroachment/evidence/boundary_evidence.json
/content/drive/MyDrive/ENVA/Agricultural_Encroachment/evidence/carbon_loss_evidence.json
/content/drive/MyDrive/ENVA/Agricultural_Encroachment/evidence/hotspot_evidence.json
/content/drive/MyDrive/ENVA/Agricultural_Encroachment/evidence/map_manifest.json
/content/drive/MyDrive/ENVA/Agricultural_Encroachment/maps/encroachment_before_after_map.html

Disclaimer: This document reports satellite-derived analytical evidence for decision support. It does not replace field surveys, cadastral records, legal review, or regulatory approval.